

QUINULT 4 NE, WA, USA

WMO: 742030

Lat: 47.5139N

Lon: 123.8120W

Elev: 286

StdP: 14.54

Time Zone: -8.00 (NAP)

Period: 06-19

WBAN: 04237

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	25.8	28.0	19.7	15.0	35.3	24.8	19.1	31.7	6.3	39.7	5.0	40.6	0.7	N/A	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.7	83.5	65.9	78.9	64.6	75.0	63.0	68.0	81.1	66.2	77.7	64.4	74.1	2.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.1	87.4	72.0	61.7	83.0	69.8	60.4	79.2	67.8	32.2	80.7	30.9	77.7	29.6	74.0	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.7	4.2	3.2	DB	21.9	92.9	3.3	3.5	19.5	95.4	17.5	97.5	15.6	99.4	13.2	102.0	(4)
(5)			WB	22.8	71.8	3.2	1.0	20.6	72.5	18.7	73.1	16.9	73.7	14.6	74.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	39.3	41.2	43.2	46.8	53.0	57.3	62.2	63.1	58.6	50.7	43.4	38.5		
	DBStd	10.02	5.42	5.12	5.10	5.59	5.72	5.30	4.37	4.67	4.98	4.30	5.75	5.42		
	HDD50	1566	332	247	219	128	30	2	0	0	2	41	207	358		
	HDD65	5617	797	665	676	548	374	242	111	93	200	442	647	822		
	CDD50	1507	1	2	8	31	124	220	377	408	261	64	10	1		
	CDD65	81	0	0	0	1	2	10	24	35	9	0	0	0		
	CDH74	1085	0	0	3	20	75	142	336	388	120	1	0	0		
	CDH80	319	0	0	0	5	13	42	104	124	31	0	0	0		
	Wind	WSAvg	1.2	1.2	1.3	1.4	1.4	1.2	1.1	1.1	1.0	1.1	1.4	1.3		
	Precipitation	PrecAvg	140.6	21.1	15.9	15.8	9.6	6.1	4.0	3.1	6.1	13.1	18.6	23.3		
		PrecMax	174.6	32.3	28.9	29.9	16.9	11.1	8.3	9.4	11.1	14.6	27.4	35.9		
		PrecMin	116.7	9.7	8.0	3.4	3.6	2.0	0.8	0.6	0.2	1.3	6.9	11.2		
		PrecStd	20.6	6.0	6.5	6.6	4.1	2.6	2.3	2.8	3.0	3.9	8.0	6.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.0	58.9	69.0	75.9	81.8	86.6	89.9	90.2	84.6	70.9	57.9	53.2		
		MCWB	51.7	51.2	53.2	57.9	61.0	69.4	67.2	67.8	65.9	58.0	56.4	51.9		
	2%	DB	50.5	54.1	59.4	66.7	74.8	77.6	81.9	83.1	76.9	64.1	55.5	50.1		
		MCWB	49.6	50.1	49.3	54.6	60.0	64.5	66.1	66.7	63.5	57.2	54.3	49.1		
	5%	DB	48.7	51.0	54.0	60.1	68.6	72.1	77.1	77.9	71.8	60.1	53.9	48.0		
		MCWB	47.8	49.0	48.0	51.3	58.6	61.9	64.8	65.2	62.0	55.9	53.1	47.3		
	10%	DB	47.0	48.8	50.6	55.4	63.8	67.9	72.9	73.3	67.1	57.3	51.8	46.0		
		MCWB	46.3	47.4	47.7	49.3	55.7	59.5	62.7	63.2	60.3	54.8	50.8	45.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.5	54.5	54.8	60.8	63.9	70.2	68.7	69.8	69.2	63.0	57.2	53.0		
		MCDB	53.1	56.0	63.3	75.2	77.1	84.1	82.3	85.9	82.5	68.2	57.6	53.6		
	2%	WB	50.4	51.4	51.8	56.7	61.3	66.8	66.9	67.7	65.5	59.3	54.9	49.7		
		MCDB	51.0	52.9	58.1	66.4	72.8	78.2	79.8	81.0	74.6	61.8	55.3	50.1		
	5%	WB	48.3	49.8	50.3	53.1	59.1	63.6	65.2	66.1	63.6	57.0	53.3	47.7		
		MCDB	48.8	51.1	54.7	59.5	69.8	72.1	77.1	77.6	70.7	58.8	54.0	48.1		
	10%	WB	46.4	48.1	48.5	50.8	57.0	60.9	63.3	64.2	61.5	55.3	51.1	45.3		
		MCDB	47.0	49.5	51.6	55.0	64.2	67.7	72.8	73.5	66.3	56.9	51.9	45.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	11.0	13.4	15.5	18.2	18.4	21.1	20.7	17.8	13.9	9.8	7.9		
		MCDBR	9.8	15.2	21.7	25.5	28.5	27.4	29.4	29.7	27.0	19.6	11.3	8.6		
		MCWBR	9.4	9.9	12.5	13.9	14.9	14.6	14.5	14.2	14.2	12.4	9.6	7.8		
		MCDBR	9.7	12.0	17.5	22.8	26.5	26.2	27.3	27.5	23.1	15.5	9.9	8.2		
	5% WB	MCWBR	9.0	9.3	10.8	13.0	14.5	14.5	14.1	13.6	12.8	11.1	9.0	7.9		
(40) Clear-Sky Solar Irradiance	taub		0.326	0.324	0.338	0.371	0.403	0.376	0.360	0.362	0.348	0.339	0.335	0.324		
	taud		2.463	2.505	2.482	2.360	2.277	2.391	2.465	2.460	2.500	2.518	2.443	2.423		
	Ebn at Noon		239	268	281	280	273	281	285	280	274	259	232	223		
	Edh at Noon		21	25	29	36	41	36	34	32	28	24	21	20		
(44) All-Sky Solar Radiation		RadAvg	317	596	903	1326	1651	1804	1950	1669	1248	712	377	263		
(45)		RadStd	44	77	101	122	135	173	149	118	111	76	49	36		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page